# GRE Quantitative Reasoning

## 모의고사 Set 5 — 문제지

**시범 세트 · S2 상향**

**구성**
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

**대소비교(Quantitative Comparison) 선택지** — 모든 QC 문제 공통
(A) Quantity A is greater.  (B) Quantity B is greater.
(C) The two quantities are equal.  (D) The relationship cannot be determined from the information given.

**안내**

- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 5 해설지)에 있습니다.

**답안 기록표**

| S1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 답 | | | | | | | | | | | | |

| S2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 답 | | | | | | | | | | | | | | | |

## Section 1 12 Questions · 21 Minutes

### Question 1 대소비교

1318. In the rectangular coordinate system, the distance between the points (a,2) and (b,6) is 5.
Quantity A: |a−b|
Quantity B: 3

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

### Question 2 대소비교

787. Quantity A: $9^{17}/8^{17}$
Quantity B: $(9^{17}+5^{10})/(8^{17}+5^{10})$

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

### Question 3 대소비교

897. Each copy of Magazine X is sold either at the cover price of $3.25 per copy or at the subscription price of $1.75 per copy. The average (arithmetic mean) price of the copies of Magazine X that were sold last month was $2.25 per copy.
Quantity A: The number of copies of Magazine X that were sold last month at the subscription price
Quantity B: Twice the number of copies of Magazine X that were sold last month at the cover price

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

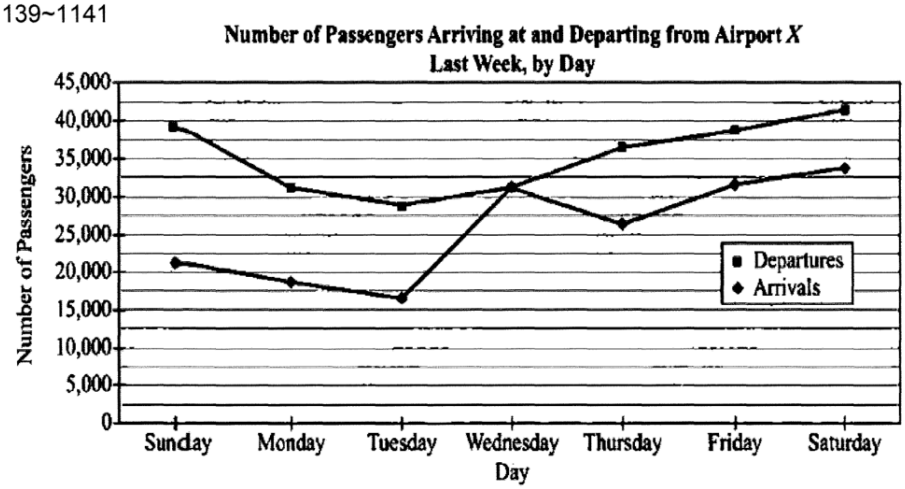
### Question 4 대소비교

740. Quantity A: The number of positive integers less than 50 that are the product of 2 different prime numbers greater than 2
Quantity B: 5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

## Question 5 객관식

1139~1141




1139. The number of passengers arriving on Sunday is approximately what fraction of the number of passengers arriving on Friday?

A1/2
B2/3
C4/5
D5/6
E6/7

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

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Questions 6 to 8 are based on the following data.

**Population and Per Capita Carbon Dioxide Emissions for Selected Countries, 2001**

| Country | Population in 2001 (millions) | Per Capita Carbon Dioxide Emissions in 2001 (metric tons) | Percent Change in Per Capita Carbon Dioxide Emissions from 1990 to 2001 |
|---|---|---|---|
| Canada | 31 | 17.1 | +10.3% |
| France | 59 | 6.2 | +1.6% |
| Germany | 82 | 10.1 | −16.6% |
| Japan | 127 | 9.1 | +10.3% |
| Mexico | 99 | 3.7 | +3.5% |
| United Kingdom | 60 | 8.9 | −8.6% |
| United States | 284 | 20.6 | +6.6% |

## Question 6 객관식

48. In 2001, what was the range of the per capita carbon dioxide emissions, in metric tons, for the countries listed?

A. 3.5
B. 9.1
C. 12.1
D. 14.4
E. 16.9

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## Question 7 객관식

49. For how many of the countries listed were the per capita carbon dioxide emissions lower in 1990 than in 2001?

A. 2
B. 3
C. 4
D. 5
E. 6

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## Question 8 객관식

50. In Canada's population increased by 3 million from 1990 to 2001, which of the following is closest to the increase in total carbon dioxide emissions, in metric tons, for Canada from 1990 to 2001?

A. 100,000
B. 1,000,000
C. 10,000,000
D. 100,000,000
E. 1,000,000,000

※ 문제 첫 단어 "In"은 "If"의 오타입니다.

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## Question 9 객관식

797. The prizes for a certain contest are in 5 sealed envelopes: 2 containing cash and the other 3 containing gift certificates. If 2 envelopes are to be randomly selected from the 5 envelopes, one at a time without replacement, what is the probability that at least one of the envelopes selected will contain a cash prize?
A1/5
B2/5
C3/5
D3/10
E7/10

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

## Question 10 주관식

1300. As a part of an environmental study of a river, a random sample of trout was drawn from the river and the lengths of the trout were recorded. The average (arithmetic mean) length was 14.31 inches. If a length of 16.89 inches was 1.50 standard deviations above the average, what was the standard deviation of the lengths of the trout in the sample?

## Question 11 복수선택

417. If x is the average (arithmetic mean) of 10 and y, and y is the average of x and 22, which of the following statements are true? Indicate all such statements.
Ax < y
By-10=22-x
CThe average of x and y is 16.

Indicate all that apply. A B C

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## Question 12 객관식

986. Of the students at a certain university, 40 percent are graduate students, and the rest are undergraduate students. If 15 percent of the graduate students and 30 percent of the undergraduate students are pursuing an engineering degree, what percent of the students pursuing an engineering degree are graduate students?
A6%
B160/3%
C20%
D25%
E331/3

※ 원본 선택지 표기 오류: (B) 160/3% → 16⅔%, (E) 331/3 → 33⅓% 로 읽으세요.

A B C D E

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## Section 2 15 Questions · 26 Minutes

### Question 1 대소비교

1110. Nancy invested $5,000 for x years at an annual interest rate of 6 percent, compounded annually.
David invested $6,000 for y years at an annual interest rate of 6 percent, compounded semiannually.
Quantity A: The total amount of interest that Nancy's investment earned
Quantity B: The total amount of interest that David's investment earned

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

### Question 2 대소비교

1055. In a club composed of 30 members, 2/5 of the members are older than 24 years and 5/6 of the members are right-handed.
Quantity A: The number of members in the club who are older than 24 years and right-handed
Quantity B: 10

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

### Question 3 대소비교

1020. In City N, the sales tax is 8 percent of the selling price. For a certain week, the sales tax was suspended on items of clothing, but only on those with selling prices under $100. During that week, a customer bought 8 items of clothing with a total selling price of $292.
Quantity A: The total amount of sales tax that was suspended on the 8 items of clothing
Quantity B: $20

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

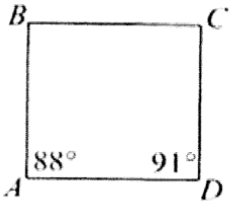
## Question 4 대소비교

801. 0<a<b<c
The average of a, b, and c is b.
Quantity A: The average (arithmetic mean) of $a^2$, $b^2$ and $c^2$
Quantity B: The median of $a^2$, $b^2$ and $c^2$

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

## Question 5 대소비교

1044.




$AD \parallel BC$

Quantity A: The length of line segment AB
Quantity B: The length of line segment DC

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

## Question 6 객관식

1072. How many distinct solutions does the equation $|1-|x-215||=1$ have?
ANone
BOne
CTwo
DThree
EFour

(A) (B) (C) (D) (E)

## Question 7 객관식

1143. If $x^2=b$ and $y^2=q$, where x and y are greater than 0, which of the following is equal to $(x+y)^2$?
A b+q
B b+2bq+q
C $b^2+2bq+q^2$
D $b+(\sqrt{bq})+q$
E $b+(2\sqrt{bq})+q$

(A) (B) (C) (D) (E)

## Question 8 객관식

1164. A certain device is a rectangular box with dimensions 20 inches by 5 inches by 6 inches. If the device has mass 17 kilograms, then the device's density is approximately how many kilograms per cubic foot? (Density is the ratio of mass to volume, and 1 foot=12 inches.)
A.20
B.30
C.40
D.50
E.60

A B C D E

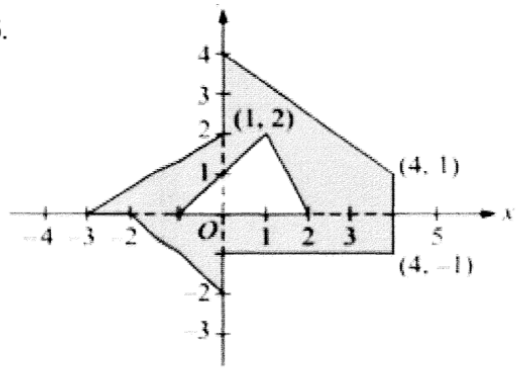
## Question 9 복수선택

330. A coffee shop sells its own brand of ground coffee packaged in two sizes-small and large. The small size is priced at k dollars per package, and the large size is priced at m dollars per package. If the total price of 4 packages of the coffee is $30, which of the following could be the values of k and m?
Indicate all such values.
Ak=4 and m=6
Bk=4 and m=16
Ck=5 and m=10
Dk=5 and m=15
Ek=7 and m=8

Indicate all that apply. A B C D E

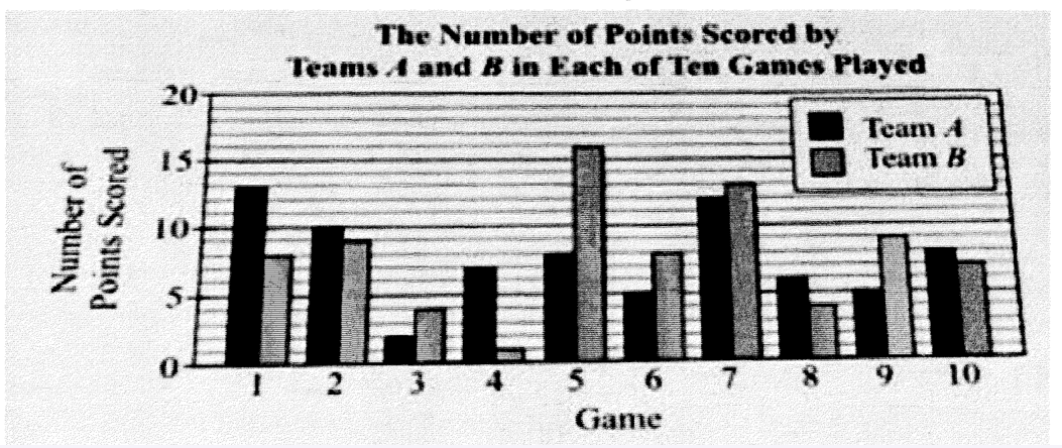
## Question 10 주관식

775.




What is the area of the shaded region shown in the xy-plane?

Questions 11 to 13 are based on the following data.




**Question 11** 객관식

57. In what percent of the games for which the difference in the number of points scored by the two teams was equal to 1 point did team A score more points than team B?

A. 20%

B. 30%

C. $33\frac{1}{3}\%$

D. 50%

E. $66\frac{2}{3}\%$

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

**Question 12** 객관식

58. Two of the ten games will be selected at random, without replacement. What is the probability that both of the games selected will be games in which team A scored more points than team B?

A. $\frac{2}{9}$

B. $\frac{1}{4}$

C. $\frac{5}{18}$

D. $\frac{1}{3}$

E. $\frac{1}{2}$

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## Question 13 객관식

59. The number of games in which team B scored at least 50 percent more points than team A was what fraction of the number of games in which team B scored more points than team A?

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{3}{5}$

E. $\frac{4}{5}$

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

## Question 14 객관식

387. Three numbers are to be selected at random and without replacement from the five numbers 4, 5, 7, 8 and 11. What is the probability that the three numbers selected could be the lengths of the sides of a triangle?
A1/5
B2/5
C3/5
D4/5
E1

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

## Question 15 복수선택

572. In a list of 25 different numbers, the average (arithmetic mean) of all the numbers greater than the median is 200, and the average of all the numbers less than the median is 100. If the average of the 25 numbers is an integer, which of the following could be the value of the median?
Indicate all such values.
A120
B125
C140
D145
E160
F175
G180

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G]